

LAB ASSIGNMENT-15.1

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Batch-46

Task 1: Student Records API

Prompt

Generate a Flask REST API for managing student records with CRUD operations using an in-memory dictionary. Include endpoints for GET, POST, PUT, DELETE and return JSON responses. Code

```
from flask import Flask, request, jsonify

app = Flask(__name__)

students = {}

# GET all students
@app.route('/students', methods=['GET']) def
get_students():
    return jsonify(students)

# POST new student
@app.route('/students', methods=['POST'])
def add_student():    data = request.json
    student_id = str(len(students) + 1)
    students[student_id] = data
    return jsonify({"message": "Student added", "id": student_id})

# PUT update student
@app.route('/students/<id>', methods=['PUT'])
def update_student(id):    if id in students:
    students[id] = request.json    return
    jsonify({"message": "Updated"})
    return jsonify({"error": "Student not found"}), 404

# DELETE student
@app.route('/students/<id>', methods=['DELETE'])
```

```
def delete_student(id):
    if id in students:
        del students[id]
        return jsonify({"message": "Deleted"})
    return jsonify({"error": "Student not found"}), 404
```

```
if __name__ ==
'__main__':
    app.run(debug=True) Test
```

Cases

```
assert isinstance(students, dict)
students["1"] = {"name": "Akhil"}
assert students["1"]["name"] ==
"Akhil" del students["1"] assert "1" not
in students
```

Task 2: Library Book API

Prompt

Create a Flask API for managing books with CRUD operations and PATCH support.

Include error handling for invalid IDs. Code

```
books = {}
```

```
@app.route('/books', methods=['GET'])
def get_books():
    return jsonify(books)
```

```
@app.route('/books', methods=['POST'])
def add_book():
    data = request.json
    book_id = str(len(books) + 1)
    books[book_id] = data
    return jsonify({"id": book_id})
```

```
@app.route('/books/<id>',
methods=['GET']) def get_book(id):    if id
in books:
    return jsonify(books[id])
    return jsonify({"error": "Not found"}), 404
```

```

@app.route('/books/<id>', methods=['PATCH'])
def update_book(id):
    if id in books:
        books[id].update(request.json)
    return jsonify({"message": "Updated"})
    return jsonify({"error": "Not found"}), 404

```

```

@app.route('/books/<id>',
methods=['DELETE']) def delete_book(id):
    if id in books:
        del books[id]
        return jsonify({"message": "Deleted"})
    return jsonify({"error": "Not found"}), 404

```

Test Cases

```

books["1"] = {"title": "Python"}
assert books["1"]["title"] == "Python"
books["1"].update({"available":
True}) assert books["1"]["available"]
== True del books["1"] assert "1" not
in books

```

Task 3: Employee Payroll API

Prompt

Build a Flask API to manage employees and salaries with endpoints for listing employees, adding employees, updating salary, and deleting employees. Add docstrings.

Code

```

employees = {}

```

```

@app.route('/employees', methods=['GET'])
def get_employees():
    """Return all employees"""
    return jsonify(employees)

```

```

@app.route('/employees',
methods=['POST']) def add_employee():
    """Add new employee"""
    data = request.json
    emp_id = str(len(employees)
+ 1)
    employees[emp_id] = data
    return jsonify({"id": emp_id})

```

```

@app.route('/employees/<id>/salary', methods=['PUT'])
def update_salary(id):    """Update employee salary"""
    if id in employees:
        employees[id]['salary'] =
        request.json.get('salary')    return
    jsonify({"message": "Salary updated"})    return
    jsonify({"error": "Not found"}), 404

```

```

@app.route('/employees/<id>',
methods=['DELETE']) def delete_employee(id):
    """Delete employee"""    if id in employees:
    del employees[id]
        return jsonify({"message": "Deleted"})
    return jsonify({"error": "Not found"}), 404

```

Test Cases

```

employees["1"] = {"name": "John", "salary":
5000} assert employees["1"]["salary"] == 5000
employees["1"]["salary"] = 6000 assert
employees["1"]["salary"] == 6000 del
employees["1"] assert "1" not in employees

```

Task 4: Online Food Ordering API

Prompt

Generate a Flask REST API for a food ordering system with endpoints for menu listing, placing orders, tracking, updating, and deleting orders. Suggest improvements.

Code

```

menu = {"1": "Pizza", "2": "Burger"}
orders = {}

```

```

@app.route('/menu', methods=['GET']) def
get_menu():
    return jsonify(menu)

```

```

@app.route('/order', methods=['POST'])
def place_order():    data = request.json
    order_id = str(len(orders) + 1)
    orders[order_id] = data
    return jsonify({"order_id": order_id})

```

```

@app.route('/order/<id>', methods=['GET'])
def get_order(id):    if id in orders:
    return jsonify(orders[id])
    return jsonify({"error": "Not found"}), 404

```

```

@app.route('/order/<id>', methods=['PUT'])
def update_order(id):    if id in orders:
    orders[id] = request.json    return
    jsonify({"message": "Updated"})
    return jsonify({"error": "Not found"}), 404

```

```

@app.route('/order/<id>',
methods=['DELETE']) def delete_order(id):
if id in orders:    del orders[id]    return
    jsonify({"message": "Cancelled"})    return
    jsonify({"error": "Not found"}), 404 Test
Cases

```

```

orders["1"] = {"item": "Pizza"} assert
orders["1"]["item"] == "Pizza"
orders["1"] = {"item": "Burger"}
assert orders["1"]["item"] ==
"Burger" del orders["1"] assert "1"
not in orders

```